21. A variable which can assume finite or countably infinite number of values is known as

(a) continuous (b) discrete

(c) qualitative (d) none of them

22. The probability function of a random variable is defined as

$X = x$	-1	-2	0	1	2
$P(x)$	k	2k	3k	4k	5k

Then k is equal to

(a) zero (b) $\frac{1}{4}$

(c) $\frac{1}{15}$ (d) one

23. If $p(x)=\frac{1}{10}$, x = 10, then $E(X)$ is

(a) zero (b) $\frac{6}{8}$

(c) 1 (d) –1

24. A discrete probability function $p(x)$ is always

(a) non-negative (b) negative

(c) one (d) zero

25. In a discrete probability distribution the sum of all the probabilities is always equal to

(a) zero

(b) one

(c) minimum

(d) maximum

26. An expected value of a random variable is equal to it's

(a) variance

(b) standard deviation

(c) mean

(d) covariance

27. A discrete probability function $p(x)$ is always non-negative and always lies between

(a) 0 and ∞ (b) 0 and 1

(c) –1 and +1 (d) –∞ and +∞

28. The probability density function $p(x)$ cannot exceed

(a) zero (b) one

(c) mean (d) infinity

29. The height of persons in a country is a random variable of the type

(a) discrete random variable

(b) continuous random variable

(c) both (a) and (b)

(d) neither (a) nor (b)

30. The distribution function $F(x)$ is equal to

(a) $P(X=x)$

(b) $P(X\leq x)$

(c) $P(X\geq x)$

(d) all of these

Miscellaneous Problems

1. The probability function of a random variable X is given by

$$p(x)=\begin{cases}\frac{1}{4}, \text{ for } x=-2\\ \frac{1}{4}, \text{ for } x=0\\ \frac{1}{2}, \text{ for } x=10\\ 0, \text{ elsewhere}\end{cases}$$

Evaluate the following probabilities.

(i) $P(X\leq 0)$ (ii) $P(X<0)$

(iii) $P(|X|\leq 2)$ (iv) $P(0\leq X\leq 10)$

2. Let X be a random variable with cumulative distribution function

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